

Case Pilot

AI surgical case intake + BlackArmor implant configuration assistant.

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Portfolio piece for icotec medical

OPEN THE LIVE DEMO

scaffold-production-3492.up.railway.app

Source code github.com/smartpneucontact-sketch/Scaffold

Model Anthropic Claude Sonnet 4.6 · ~\$0.06 per case

SUMMARY

Case Pilot is a working portfolio demo aligned to the icotec Digitalization & AI Professional role. A surgeon's free-text case description goes in, a structured case record comes out — cited indication checks, recommended BlackArmor implant configuration, imaging-benefit rating, MDR / FDA compliance flags. The agent loop and corpus contract map cleanly onto a Microsoft Power Platform + Azure AI production deployment.

THE PROBLEM THIS SOLVES

Surgical case intake at a medical-device company today is mostly paper or free-text email. A clinical specialist reads the surgeon's description, manually checks indication alignment against the IFU, picks an implant configuration from the product catalog, flags any off-label use, and produces a structured record for the OR and the regulatory trail. High-frequency, citation-heavy, document-bound — the textbook 'paper-to-digital with AI' workflow icotec's JD explicitly names.

WHAT IT SHIPS

- Single AI agent that runs a tool-use loop on Claude Sonnet 4.6: retrieve corpus, check imaging benefit, classify case complexity, then draft the structured record.
- Synthetic but realistic icotec corpus: BlackArmor device specification, Indications for Use, surgical technique guide, prior cases, regulatory & quality summary.
- FastAPI service with visitor-notify (Resend HTTPS, dedup + bot filter), modern dark UI with semantic rendering of every output field, deployed to Railway.
- The same shape ports to a Power Platform deploy: Power Apps canvas UI, Power Automate flow → Azure Function custom connector, Azure AI Search over the SharePoint document library, Dataverse for implant inventory, App Insights for telemetry.

HOW IT MAPS TO ICOTEC'S STACK

The agent loop and the corpus contract are the durable part. The surfaces around them change to fit icotec's Microsoft-first environment.

THIS DEMO	ICOTEC PRODUCTION TARGET
Anthropic SDK + Claude Sonnet 4.6	Azure OpenAI via Power Platform AI Builder
BM25 over markdown corpus	Azure AI Search / SharePoint Search
FastAPI + Pydantic backend	Power Automate flow + custom connector to Azure Function
Single-page HTML UI	Power Apps canvas app, embedded in Dynamics 365 / Teams
JSONL traces on disk	App Insights → Azure Data Explorer / Fabric
Synthetic corpus	icotec IFU + surgical guides in SharePoint + Dataverse
Visitor-notify webhook	Power Automate flow → Outlook / Teams alert
GitHub Actions CI	Same

SAMPLE AGENT REASONING (REAL RUN)

Tool sequence for an oncology case: 55F with T8 metastatic breast carcinoma, planned post-op radiation. Every retrieval and classification is captured in the trace.

RUN_ABCD1234EF56 · 3 STEPS · ONCOLOGY CASE

```
step 0  retrieve(query="BlackArmor pedicle screw indications oncology")
        -> indications:blackarmor_pedicle_screws_ifu  (score 12.4)
        -> case:case_001                             (score 8.9)

step 0  retrieve(query="radiation planning imaging artifact titanium",
                 source_type="surgical_technique")
        -> surgical_technique:blackarmor_pedicle_screw_technique  (score 7.1)

step 0  check_imaging_benefit(oncology=true, post_op_radiation=true,
                              post_op_mri_planned=true)
        -> benefit_band="high", score=5

step 1  classify_case_complexity(levels=5, revision=false,
                                osteoporosis=false, deformity=false)
        -> complexity_band="medium"

step 2  final answer (cited): T6-T10 fusion, BlackArmor 6.0x45 mm screws,
        CFR-PEEK rods, on-label, no specialist review required.
```

WHY THIS FITS THE ROLE

icotec's JD describes someone who turns paper processes into digital ones, builds quick wins with low-code, applies AI to streamline workflows, and runs adoption. Case Pilot is exactly that artifact: a paper-to-structured-record workflow built in days, with a clear production path onto the Microsoft stack icotec already uses. The agent pattern transfers — the next workflow could be adverse-event triage,

complaint handling, regulatory submission drafting, or sales-rep training-completion checks. Same loop, different corpus, weeks not quarters.

HONEST GAPS

- Synthetic corpus, not real IFUs. Real deployment needs PDF OCR + table extraction.
- No Dataverse / Power Platform integration yet — this is the FastAPI prototype.
- No PHI or clinical-grade safety story. This is a planning aid, not regulated decision support.
- No live UDI lookup. Production would call Dataverse implant inventory in real time.

IN ONE LINE

Paper-to-digital, AI-powered, icotec-shaped — shippable to a real workflow in a sprint on the Microsoft stack icotec already runs on.